

DISCRETE STRUCTURES I

Rules of Inferences

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1 Introduction

Proofs in mathematics are valid arguments that establish the truth of mathematical statements. By an argument, we mean a sequence of statements that end with a conclusion. By valid, we mean that the conclusion, or final statement of the argument, must follow from the truth of the preceding statements, or premises, of the argument. That is, an argument is valid if and only if it is impossible for all the premises to be true and the conclusion to be false. To deduce new statements from statements we already have, we use rules of inference which are templates for constructing valid arguments. Rules of inference are our basic tools for establishing the truth of statements.

2 Valid Arguments in Propositional Logic

Definition 1. An argument in propositional logic is a sequence of propositions. All but the final proposition in the argument are called premises and the final proposition is called the conclusion. An argument is valid if the truth of all its premises implies that the conclusion is true. An argument form in propositional logic is a sequence of compound propositions involving propositional variables. An argument form is valid no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if the premises are all true.

From the definition of a valid argument form we see that the argument form with premises $p_1, p_2, \dots, p_n$ and conclusion q is valid, when $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is a tautology.

3 Rules of Inference for Propositional Logic

We can always use a truth table to show that an argument form is valid. We do this by showing that whenever the premises are true, the conclusion must also be true. However, this can be a tedious approach. For example, when an argument form involves 10 different propositional variables, to use a truth table to show this argument form is valid requires $2^{10} = 1024$ different rows. Fortunately, we do not have to resort to truth tables. Instead, we can first establish the validity of some relatively simple argument forms, called rules of inference. These rules of inference

can be used as building blocks to construct more complicated valid argument forms. We will now introduce the most important rules of inference in propositional logic.

TABLE 1 Rules of Inference.		
Rule of Inference	Tautology	Name
$\frac{p \quad p \rightarrow q}{\therefore q}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\frac{\neg q \quad p \rightarrow q}{\therefore \neg p}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\frac{p \vee q \quad \neg p}{\therefore q}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
$\frac{p}{\therefore p \vee q}$	$p \rightarrow (p \vee q)$	Addition
$\frac{p \wedge q}{\therefore p}$	$(p \wedge q) \rightarrow p$	Simplification
$\frac{p \quad q}{\therefore p \wedge q}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\frac{p \vee q \quad \neg p \vee r}{\therefore q \vee r}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

Example : Show that the premises "It is not sunny this afternoon and it is colder than yesterday", "We will go swimming only if it is sunny", "If we do not go swimming, then we will take a canoe trip" and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset".

Let p be the proposition "It is sunny this afternoon", q the proposition "It is colder than yesterday", r the proposition "We will go swimming", s the proposition "We will take a canoe trip", and t the proposition "We will be home by sunset". Then the premises become $\neg p \wedge q$, $r \rightarrow p$, $\neg r \rightarrow s$, and $s \rightarrow t$. The conclusion is simply t . We need to give a valid argument with premises $\neg p \wedge q$, $r \rightarrow p$, $\neg r \rightarrow s$, and $s \rightarrow t$ and conclusion t .

We construct an argument to show that our premises lead to the desired conclusion as follows.

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \longrightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \longrightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \longrightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

4 Exercises

Exercise 1 :

Using rules of inferences show the following argument is valid.

1. $p \longrightarrow (q \longrightarrow r)$
 2. $p \vee s$
 3. $r \longrightarrow p$
 4. $t \longrightarrow q$
 5. $\neg s$
-
- $\therefore \neg r \longrightarrow \neg t$

Exercise 2 :

Show taht the following argument is valid or not using truth table

1. $p \wedge q$
 2. $p \longrightarrow r \wedge q$
 3. $r \longrightarrow p$
-
- $\therefore \neg r$

Exercise 3 :

If my golden ring is on the table, then it is neither beside the bathroom mirror nor in the jewelry box. My golden ring is not in my finger and it is not in my bag. If I did not lose my golden ring, then it is in my bag. My golden ring is in my finger or in the jewelry box. Therefore, my golden ring is not on the table and I lost it.

1. Convert the above argument into symbolic form.
2. Using the inference rules, show the above argument is valid.

Exercise 4 :

The red ball is under the first cup or it is not under the third cup. If the red ball is not under the second cup or it is under the fourth cup then the red ball is not under the first cup. The red ball is under the third cup but it is not under the fifth cup. Therefore, the red ball is under the second cup.

1. Convert the above argument into symbolic form.
2. Using the inference rules, show the above argument is valid.

Exercise 5 :

Show the argument is invalid using truth assignment.

- $$\begin{array}{l}
 p \longrightarrow q \\
 r \vee q \\
 \neg s \longrightarrow \neg t \\
 \neg q \vee s \\
 \neg s \wedge \neg q \\
 w \vee t \\
 \neg p \wedge r \longrightarrow u \\
 \hline
 \therefore \neg u \wedge \neg w
 \end{array}$$